

PHAYAO, Thailand

WMO: 483100

Lat: 19.1931N

Lon: 99.8836E

Elev: 1309

StdP: 14.01

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	52.7	55.2	47.6	51.2	70.6	50.5	57.1	67.9	9.1	79.0	7.6	78.4	0.0	320	0.270

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	20.8	98.7	74.1	96.5	74.4	94.4	74.6	80.6	89.8	79.8	88.8	79.2	87.9	2.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.3	154.2	85.1	77.4	149.9	84.2	76.9	147.2	83.7	45.5	89.8	44.7	89.2	44.0	88.1	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.9	8.1	6.9	DB	48.1	101.3	3.7	1.9	45.5	102.7	43.3	103.8	41.2	104.9	38.6	106.3	
			WB	47.3	82.4	3.7	1.5	44.6	83.5	42.5	84.3	40.4	85.2	37.7	86.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.1	71.4	75.3	80.5	84.5	83.4	83.0	81.8	81.2	81.1	79.6	75.6	71.3
	DBStd	5.48	3.67	3.65	4.11	3.88	3.75	2.31	2.38	2.02	2.03	2.66	3.71	4.50
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	12	4	0	1	0	0	0	0	0	0	0	0	7
	CDD50	10610	664	708	947	1035	1034	989	986	968	932	917	768	662
	CDD65	5147	203	288	482	585	569	539	521	503	482	452	319	204
	CDH74	49263	1521	2679	5222	6908	6206	5787	5122	4625	4225	3467	2175	1326
CDH80	20006	453	1183	2715	3698	2844	2261	1781	1475	1376	1170	712	338	
Wind	WSAvg	1.2	0.8	1.1	1.4	1.7	1.5	2.0	1.8	1.3	0.7	0.5	0.5	0.6
Precipitation	PrecAvg	46.3	0.5	0.4	1.4	3.2	7.0	4.1	6.2	8.4	8.3	4.7	1.5	0.5
	PrecMax	60.8	3.0	3.2	7.2	8.0	12.2	8.3	11.2	15.2	14.3	9.0	6.3	4.0
	PrecMin	32.5	0.0	0.0	0.0	0.4	0.6	1.1	2.2	3.7	3.4	1.1	0.0	0.0
	PrecStd	8.3	0.8	0.7	1.8	1.9	2.6	1.7	2.4	3.0	2.6	2.3	1.6	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.4	94.6	99.0	102.1	100.8	95.4	94.0	92.0	91.8	91.9	90.9	88.5
		MCWB	68.8	68.5	70.8	74.6	77.9	78.4	78.4	78.8	78.8	78.2	76.3	73.5
	2%	DB	86.9	92.2	97.1	99.5	96.9	93.0	91.5	90.1	90.2	89.7	88.4	85.9
		MCWB	69.0	68.8	70.9	74.5	77.8	78.4	78.0	78.6	78.5	77.5	75.1	72.0
	5%	DB	84.8	90.1	95.0	97.3	94.3	91.1	89.7	88.6	88.7	88.2	86.6	83.8
		MCWB	68.9	68.9	70.9	74.8	77.8	78.1	77.9	78.3	78.1	76.9	74.2	71.2
	10%	DB	82.6	87.7	92.6	95.0	91.6	89.4	88.0	87.0	87.0	86.4	84.5	81.5
		MCWB	68.4	68.7	71.1	74.9	77.7	77.9	77.7	77.9	77.8	76.5	73.3	69.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.8	75.0	78.0	79.9	81.3	81.2	81.0	80.8	81.0	80.1	78.4	76.4
		MCDB	83.2	86.7	90.3	94.1	92.9	90.3	89.0	88.4	88.5	88.3	87.3	85.4
	2%	WB	72.4	73.4	75.8	78.4	80.0	80.0	79.7	79.8	79.8	78.9	77.0	74.5
		MCDB	81.1	85.1	88.6	92.2	91.4	89.1	87.6	87.3	87.3	86.8	85.2	82.2
	5%	WB	71.0	72.0	74.5	77.5	79.1	79.2	78.8	79.0	79.0	78.0	75.8	72.8
		MCDB	79.9	83.4	87.4	90.9	90.0	88.0	86.7	86.4	86.3	85.4	83.4	80.0
	10%	WB	69.7	70.7	73.4	76.5	78.2	78.4	78.1	78.3	78.2	77.2	74.7	71.4
		MCDB	78.7	82.2	86.4	89.4	88.4	86.8	85.6	85.1	85.0	84.1	81.7	78.6
Mean Daily Temperature Range	5% DB	MDBR	25.0	27.7	25.3	20.8	15.7	13.0	12.1	11.9	12.8	14.9	19.1	22.0
		MCDBR	27.8	30.5	28.4	23.7	20.0	15.7	15.0	14.3	15.1	17.0	20.9	23.2
		MCWBR	11.5	10.6	8.5	6.5	6.2	4.9	5.0	5.1	5.3	6.1	8.4	10.0
	5% WB	MCDBR	22.8	25.4	23.2	20.6	17.0	13.9	13.3	13.2	14.1	15.3	17.9	19.9
MCWBR		10.7	10.1	8.4	7.1	6.3	5.2	5.2	5.3	5.5	6.1	7.7	9.4	
Clear-Sky Solar Irradiance	taub		0.432	0.520	0.685	0.667	0.496	0.441	0.433	0.452	0.478	0.509	0.427	0.425
	taud		2.198	1.962	1.604	1.702	2.190	2.375	2.391	2.317	2.224	2.091	2.285	2.246
	Ebn at Noon		266	248	213	220	258	270	272	269	262	248	266	265
	Edh at Noon		43	58	85	78	47	39	38	42	45	50	39	40
All-Sky Solar Radiation	RadAvg		1432	1682	1743	1877	1706	1520	1403	1389	1455	1417	1442	1343
	RadStd		78	75	103	113	151	103	100	83	62	69	100	83

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	+0.34	N/A	N/A	N/A	N/A
Regional (13 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+274	+270

Nomenclature: See separate page